END-TO-END CIVIC EVIDENCE & ACCOUNTABILITY WORKFLOWS

PARAKH

Use Case Document

How a civic problem travels from first observation to verified, traceable outcome

VERSION	1.0
DATE	7 August 2026
OWNER	Hassan — Founder & Project Lead
DERIVED FROM	Parakh Master Document v1.2.1
COMPANION TO	Parakh User Training & Operations Manual v1.0 — separate document
AUDIENCE	Government officials · institutional partners · civic-technology reviewers · researchers · evaluators · developers

CURRENT STATUS — READ FIRST

Parakh is currently under design and development. It is **not** an official Government of Punjab application. **No formal partnership** with any government department or NGO has been signed. The complete application is **not yet fully deployed**, and **no social impact has yet been demonstrated**.

This document describes two different things and always labels which is which:

PHASE 1 — capability designed for the supervised pilot. FUTURE — the intended institutional operating model, **not built and not deployed**.

Contents

Use Case Document 1

1. Purpose of This Document 3

2. Parakh in One Page 3

2.1 The four properties that make a record evidence 4

3. Actors and the Five-Portal Ecosystem 4

3.1 The operational chain 5

3.2 What each actor does — and does not 5

4. Master Use Case — One Civic Issue from 0 to 100 7

Stage 1 — Observation 7

Stage 2 — Live Evidence Capture 7

Stage 3 — AI Assistance 8

Stage 4 — Human Moderation 8

Stage 5 — Independent Community Verification 8

Stage 6 — Verified Issue 9

Stage 7 — Institutional Action 9

Stage 8 — Contractor Assignment 10

Stage 9 — Contractor Work 10

Stage 10 — Before / After Verification 10

Stage 11 — Fix Claimed 11

Stage 12 — Human Confirmation 11

Stage 13 — Citizen Confirmation or Dispute 11

Stage 14 — Resolution or Rework 11

Stage 15 — Traceable Outcome 11

5. UC-01 — Citizen Reporting 12

6. UC-02 — Street / Community Representative 13

7. UC-03 — Union Council Officer / UC Operations 14

8. UC-04 — Contractor / Thekedar 15

8.1 Defect liability and recurrence 16

9. UC-05 — Mayor & Operations Command 16

9.1 The heatmap — what it is, and what it is not 17

10. UC-06 — Human Moderation 18

11. Cross-Cutting — Privacy and Reporter Protection 18

11.1 Why this matters operationally 19

12. Minor / Student Participation 19

13. AI Versus Human Decision 20

13.1 Where the humans decide 20

14. Rework and Recurrence 21

14.1 Rework — when completion evidence is not accepted 21

14.2 Recurrence during the defect liability period 21

15. Appeal and Dispute 22

16. Important Exception Flows 22

17. The Commit–Verify–Release Boundary 23

18. The End-to-End Accountability Map 24

18.1 The chain in one column 24

18.2 What holds it together 24

1. Purpose of This Document

This document answers one question:

> When a civic problem is reported in Parakh, exactly **who does what** — from the first observation to the final verified outcome?

It describes **actor-to-actor workflows**: who acts, on what evidence, under whose authority, with what recorded afterwards. It is deliberately not a training manual and not an interface design.

This document	Companion documents
Practical workflows between actors, and how responsibility moves	**User Training & Operations Manual v1.0** — step-by-step operating instructions, exception training, quick-start guides
The shape of the accountability chain	**Master Document v1.2.1** — the authoritative design record, decision register and requirements
What is Phase 1 and what is a future institutional model	*Prototype documentation* — what must be built and demonstrated

Source of truth. Where this document and the Master Document v1.2.1 disagree, **the Master Document governs**. Nothing here adds a feature, policy, partnership, statistic, deadline, capability or legal power that is not already recorded there.

How to read it. §2–§4 explain the model and follow one complete case end to end. §5–§10 present each actor as a structured use case. §11–§17 cover the cross-cutting rules — privacy, minors, the AI boundary, rework, appeals, exceptions and the financial boundary. §18 is the whole chain on one page.

2. Parakh in One Page

> Most complaint systems digitise the **complaint**.
> Parakh is designed to digitise the **proof**.

A complaint is an **assertion** — a claim that something is wrong. Evidence is something **another person can later check**. That difference is the entire product.

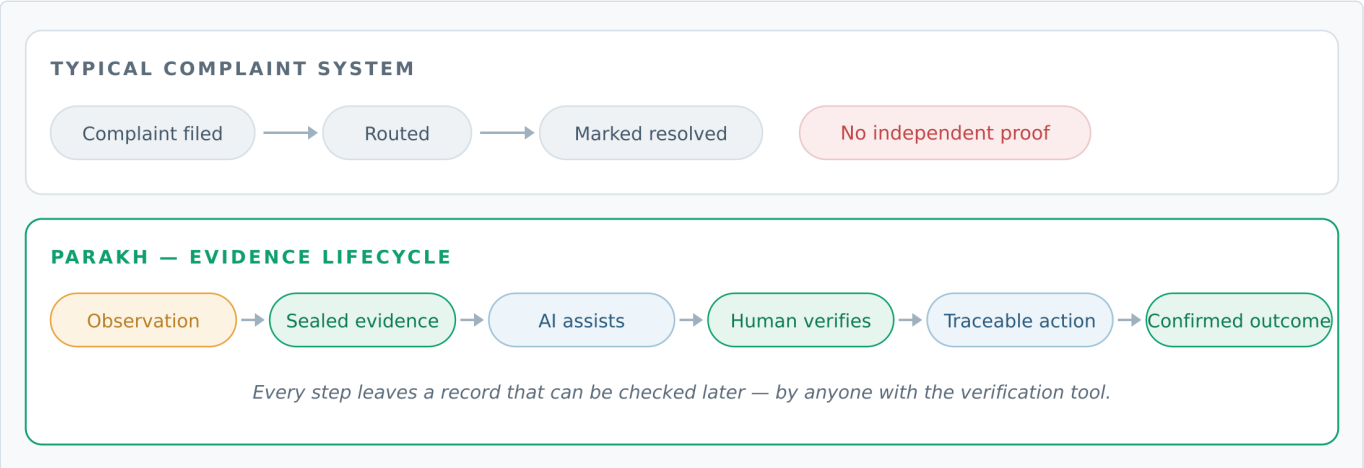


Figure 2.1 — What a conventional complaint record contains, and what a Parakh evidence record contains.

Traditional complaint flow

1. Complaint submitted
2. Routed to a department
3. Marked resolved

Nobody independently checked that the problem existed. Nobody independently checked that it was fixed. When it fails again, the trail has gone cold.

Parakh evidence lifecycle

1. Observation
2. Live evidence capture
3. AI assistance *(advisory)*
4. Human moderation
5. Independent field verification
6. Institutional action F
7. Work evidence F
8. Independent verification
9. Citizen confirmation or dispute
10. Traceable outcome

Every consequential step carries a named human, a reason and a time.

2.1 The four properties that make a record evidence

Property	How it is achieved
Captured, not uploaded	Media is taken live inside the application. There is no gallery import on the live-evidence reporting path, so an image is a claim about a specific place at a specific moment
Sealed at capture	A content hash, location and timestamp are computed at the moment of capture; storage is write-once, so later alteration is detectable
Separated from interpretation	AI transcripts, category and severity suggestions are stored as **annotations**, distinct from the original media, tagged with model and version. AI never modifies the original
Confirmed by people	A human moderator accepts it; an independent representative confirms it on the ground; the affected citizen confirms or disputes the outcome

3. Actors and the Five-Portal Ecosystem

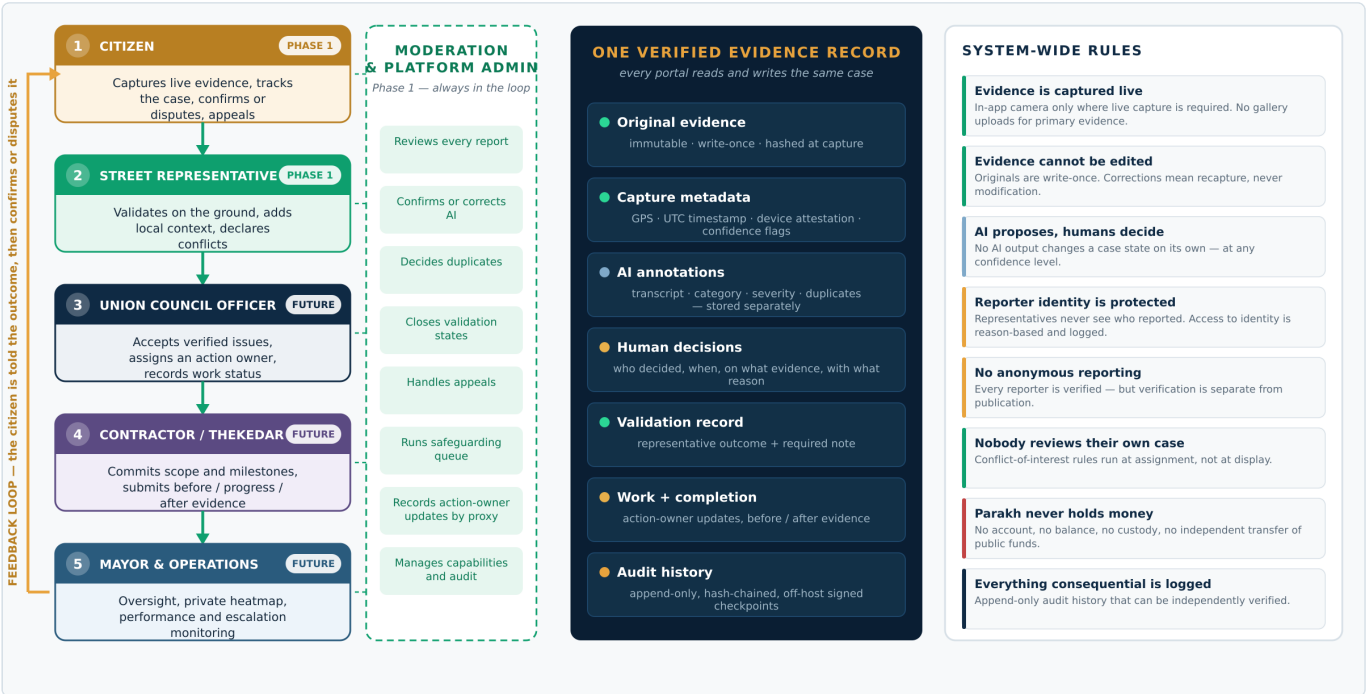


Figure 3.1 — The five operational portals, the human review layer that runs alongside them, and the single evidence record they all act on.

3.1 The operational chain

CITIZEN → STREET / COMMUNITY REPRESENTATIVE → UNION COUNCIL OFFICER / UC OPERATIONS [F] → CONTRACTOR / THEKEDAR [F] → MAYOR & OPERATIONS COMMAND [F]

…and back to the CITIZEN, who confirms or disputes the claimed outcome.

Moderation and administration are a governance layer, not a sixth public portal. A moderator sits alongside the chain: accepting reports into the workflow, closing validation states, handling appeals and safeguarding, and — in Phase 1, where no institutional actor exists — recording pilot-site authority updates by proxy under audit. Administrators hold the highest-authority actions.

3.2 What each actor does — and does not

CITIZEN PHASE 1

Responsibility · Report a real problem with live evidence, then confirm or dispute the claimed fix.

Sees · Their own case, its status and history; public-safe information about issues they support.

Does not see · Other people's identities; internal review notes; the operational heatmap.

Decides · Whether to report; whether the fix is acceptable.

Cannot · Report anonymously; upload a gallery photograph on the live-evidence path; support the same issue twice.

STREET / COMMUNITY REPRESENTATIVE PHASE 1

Responsibility · Independently confirm that a reported problem is real and observable.

Sees · The evidence packet — media, transcript, AI annotations with confidence, location, duplicate history.

Does not see · **Reporter identity, ever.** Cases outside their scope.

Decides · Recommends VALIDATED or NOT_VALIDATED, with a required note.

Cannot · Verify their own report; delete or hide anything; overrule a moderator; label a citizen fraudulent; make an institutional decision.

MODERATOR / ADMINISTRATOR PHASE 1 — GOVERNANCE LAYER

Responsibility · Human control layer: triage, duplicate decisions, state closure, appeals, safeguarding.

Sees · Everything operational. **Not** reporter identity in normal work — vault access is separately authorised, justified and logged.

Decides · Accept, reject with reason, merge duplicates, close validation states, handle appeals.

Cannot · Delete a report — no such capability exists. A **moderator may apply a temporary restriction only; permanent suspension is an administrator action**.

UNION COUNCIL OFFICER / UC OPERATIONS FUTURE

Responsibility · Turn a verified issue into an institutional action.

Sees · The full evidence bundle, validation note, supporter count, recurrence history, audit trail.

Does not see · Reporter identity.

Decides · Accept into the operations workflow or reroute; assign the action owner; record scope, cost, budget reference, timeline and milestones.

Cannot · Select the contractor; approve contracts; alter evidence or audit history.

CONTRACTOR / THEKEDAR FUTURE

Responsibility · Perform authorised work and prove it with comparable evidence.

Sees · Only their own work orders, and the before evidence for them.

Does not see · Reporter identity; other contractors' work; jurisdiction-wide data.

Decides · Whether to accept an assignment; what evidence to submit; how to answer a rejection.

Cannot · Upload gallery photographs; edit submitted evidence; confirm their own work as resolved; receive money through Parakh.

MAYOR & OPERATIONS COMMAND

FUTURE (LIMITED ADMIN FORM IN PHASE 1)

Responsibility · Situational awareness and capacity direction — not individual case handling.

Sees · Aggregated, geo-fuzzed patterns; ageing; recurrence; category trends; platform health.

Does not see · Reporter identity; precise reporter locations.

Decides · Where to direct capacity; what to escalate.

Cannot · Publish the heatmap; edit evidence or audit history; override a citizen's confirmation; approve payments.

Phase boundary. Phase 1 is a **supervised pilot** built around Citizen + Street/Community Representative + Moderator support. **Union Council Officer, Contractor, and Mayor & Operations Command are future institutional capabilities** — designed, documented, and **not built or deployed**. Where this document describes them, it describes the intended operating model, not a shipped feature.

In Phase 1 the "action owner" is the pilot-site authority — a school, NGO partner or community action group — and a moderator or administrator records their acknowledgement, work-start and fix-claimed updates by proxy, under audit. The workflow states are identical; only who holds the capability changes.

4. Master Use Case — One Civic Issue from 0 to 100

The case. After two nights of monsoon rain, a street drain in a Punjab village has collapsed and the road above it has flooded and broken open. Water stands across the width of the street outside a school. Children wade through it twice a day. **Aisha**, a shopkeeper whose children attend the school, decides to report it.

Names and times are illustrative. The stages, states, roles and rules are taken from the approved design.

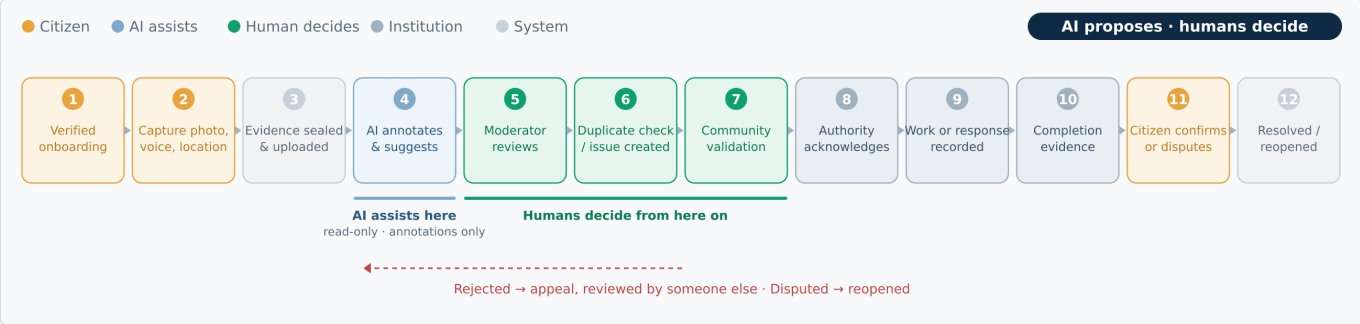


Figure 4.1 — The lifecycle this case traverses, with the human decision points marked.

Stage 1 — Observation PHASE 1

Actor	Citizen
Action	Notices a civic problem and decides to report it
Human decision	Whether this is a civic infrastructure problem at all. Parakh is not an emergency service — anything requiring immediate emergency response goes to the emergency services directly
Next	Citizen

Stage 2 — Live Evidence Capture PHASE 1

Actor	Citizen
Evidence / input	1–3 **live** photographs · GPS location · capture timestamp · a spoken description
System response	Camera opens directly — **there is no gallery import on the live-evidence reporting path**. A content hash, location and timestamp are attached at capture. Media is stored write-once
Language	**Urdu and English, including Urdu-English code-switching, at launch.** Punjabi and other regional languages are planned under Parakh's multilingual-extensible architecture and are **not a Phase-1 commitment**
AI role	None. AI never runs before capture and never touches the original
Status change	DRAFT → **SUBMITTED**
Audit	Media captured with hash; report submitted with bundle hash
Exception	Weak or absent GPS → manual map placement, flagged `location_unverified` ; the report is still accepted. No connectivity → the bundle is encrypted on the device and queued, uploading later; **capture time governs, not upload time**. Unverifiable device clock → flagged `time_unverified`
Next	The system

> **Why voice.** Literacy and digital-access barriers remain significant across Punjab, and everyday speech mixes Urdu and English within single sentences. A form that requires typing and choosing a category excludes exactly the people most affected by broken infrastructure. The core path is completable with **camera and voice alone**.

Stage 3 — AI Assistance PHASE 1

Actor	AI services (detached worker tier)
AI role	Transcribe the voice note · suggest a category *(drainage / road damage)* · suggest a severity · propose possible duplicates · produce structured observations. Every output carries a **confidence value and a model version**
System response	All outputs stored as **annotations, separate from the original evidence**, which is untouched
Human decision	None yet — **AI does not make the consequential civic decision**
Status change	SUBMITTED → PROCESSING → **AI_PROCESSED**
Exception	Low confidence → flagged for human attention, audio remains the evidence. Model unavailable → the report proceeds to human review **without annotations**. Degraded AI never blocks a case
Next	Moderator

Stage 4 — Human Moderation PHASE 1

Actor	Moderator
Inputs	Original media · transcript · AI annotations with confidence · location · duplicate candidates. **Not** the reporter's identity
Human decision	Is this a real, actionable civic issue? Is it a duplicate? Is there a safeguarding concern?
System response	Accepted → a structured **issue** is created in **OPEN_UNVALIDATED**. Rejected → REJECTED with a recorded reason and an appeal path. Duplicate → merged; the second reporter becomes a **supporter**
Status change	AI_PROCESSED → **MODERATION_APPROVED** or **REJECTED**
Audit	Decision recorded with actor, time and reason
Exception	Safeguarding concern → immediate escalation on the restricted route, content restricted
Next	Street Representative

Stage 5 — Independent Community Verification PHASE 1

Actor	Street / Community Representative
Assignment	Scoped to their named area. Parakh checks **at assignment** that they did not report it, are not named in it, and have no declared conflict — a conflicted item never arrives
What they see	Everything about the problem. **Nothing about the reporter** — identity is removed in the query that builds the packet, not hidden by a screen setting
Action	Visit the site where it is safe to do so; check independently; record findings
Evidence	A **required note** explaining the outcome; optional supplementary live-captured photographs; local context — how long, who is affected, whether it recurs

Human decision	Recommend **VALIDATED** or **NOT_VALIDATED**. They may also corroborate, challenge, or request one specific missing detail
Cannot	Verify their own report; make an institutional government decision; delete anything
Exception	Conflict → **recuse**, logged, reassigned. Site unsafe → mark `unsafe_to_verify`, which **raises** severity rather than hiding the issue. Outside scope → reroute suggestion. Welfare concern → immediate safeguarding escalation
Next	Moderator, who closes the state

> **The representative verifies the condition, not the citizen.** Their job is to answer *"does this problem exist, at this place, roughly as described?"* — not to judge who reported it, whether it is worth fixing, or whose budget it is.

Stage 6 — Verified Issue PHASE 1

Actor	Moderator closes the state on the representative's recommendation
Status change	OPEN_UNVALIDATED → **VERIFIED** or **NOT_VALIDATED**
Meaning	A report becomes a **verified civic issue** when an independent human, who did not report it and has no declared interest in it, has confirmed the condition on the ground and a moderator has recorded that outcome
Notification	Citizen and supporters are told it is verified, or told it was not validated together with the appeal path
Parallel	Other citizens may **support** the issue — one support per verified account. Support is an input to prioritisation, never an automatic trigger
Next	Action owner

Stage 7 — Institutional Action FUTURE

Actor	Union Council Officer / UC Operations
Inputs	Full evidence bundle · transcript · AI annotations with provenance · the representative's note · verified supporter count · duplicate and recurrence history · audit trail. **Not reporter identity**
Human decision	Is this a municipal responsibility, and which unit owns it? Accept into the operations workflow, or reroute with a recorded reason
Records	Work scope · estimated or approved cost against the applicable rate card · budget reference · timeline · milestones · administrative approval per the authority's own rules
AI role	May flag cost outliers and duplicate work orders. **A flag is a prompt to look, never an accusation and never a rejection**
Status change	VERIFIED → **IN_PROGRESS** on acknowledgement or work start
Statutory caveat	Which tier or department is responsible **remains subject to institutional and legal mapping** under the current governing framework. Parakh keeps the action owner **configurable** rather than hard-coded to any office
Exception	Wrong unit → reassign; the correction stays visible in the audit trail. Late acknowledgement → the 7-day benchmark is a **proposed pilot benchmark, reported as a metric and not enforced in Phase 1**
Next	Procurement authority

Stage 8 — Contractor Assignment FUTURE

> **PARAKH DOES NOT SELECT THE CONTRACTOR.** The appropriate government or procurement authority selects the contractor under applicable rules, in its own systems. Parakh **records or links** the resulting authorised assignment where institutional integration allows it. Parakh does not approve contracts, enforce penalties, or replace any procurement process.

Stage 9 — Contractor Work FUTURE

Actor	Contractor / Thekedar
Action	Accept the authorised work order · record before-work site condition · record progress at each milestone · record any delay with a **written explanation** · raise a variation request where the work as found differs from the work as scoped
Evidence	Live-captured photographs through the **same pipeline citizens use** — the integrity guarantees do not change by role
Human decision	Variations are decided by the **authorising officer**, not by Parakh and not by AI
Sees	The **before evidence**, because completion evidence must be comparable to it
Next	Contractor submits completion evidence

Stage 10 — Before / After Verification FUTURE

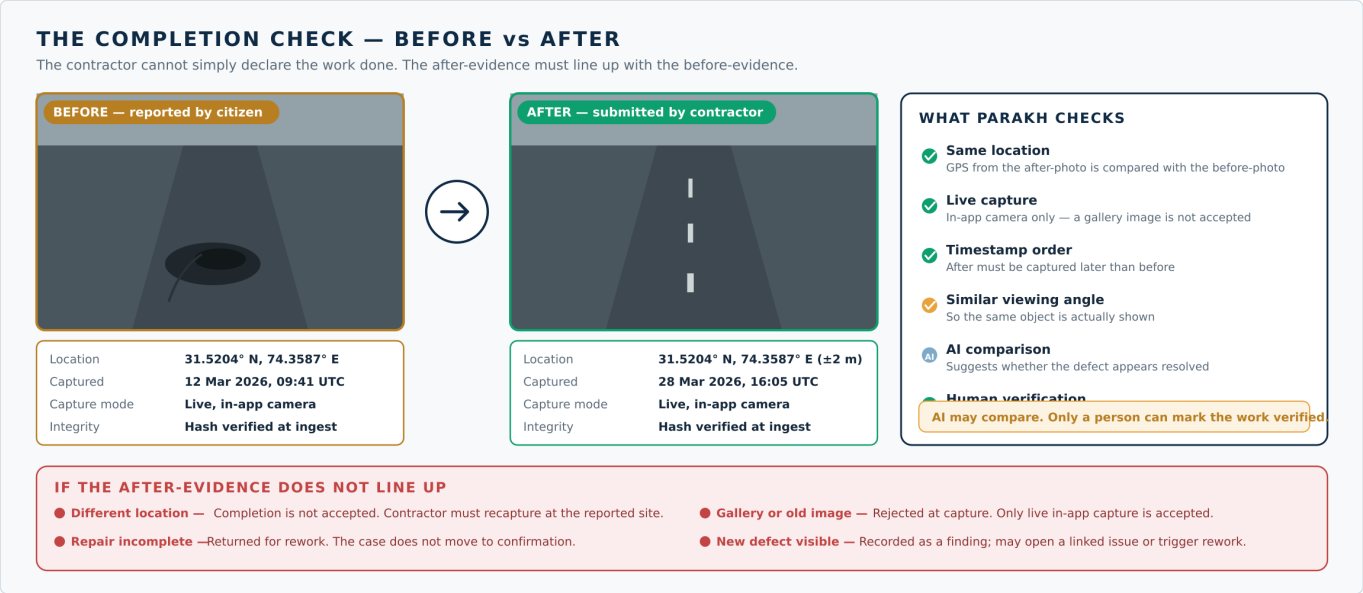


Figure 4.2 — What is checked when completion evidence is compared with the original evidence, and what happens when a check fails.

Rule	Completion evidence must be **captured live**, from the **same location** and, where the original allows it, the **same or a sufficiently similar viewing angle** as the before evidence
Checks	Live capture · GPS proximity · timestamp within the work window · viewing-angle similarity · integrity hash · metadata completeness
AI role	Compare before and after; **describe** observable differences; flag apparent mismatches. **Advisory only**
Human decision	**Mandatory.** A human decides whether the work is acceptable. AI does not approve work, does not reject work, and does not certify completion
If a check fails	The submission is **flagged for human review** — never silently accepted, never silently destroyed

Stage 11 — Fix Claimed FUTURE

> **FIX_CLAIMED ≠ RESOLVED.** A contractor cannot declare their own work independently verified. **FIX_CLAIMED** records only that someone says the work is done. It is a claim awaiting verification, and it is the point at which citizens and supporters are invited to look.

Stage 12 — Human Confirmation

Actor	Independent human reviewer — moderator, representative or authorised officer, depending on phase
Human decision	Does the evidence support completion? Complete · incomplete · not comparable · a different defect visible
Cannot	The person who submitted the evidence can never be the person who verifies it
Next	Citizen

Stage 13 — Citizen Confirmation or Dispute PHASE 1

Actor	Citizen and supporters
Action	Confirm the fix, or dispute it with a reason and optional fresh live-captured evidence
Human decision	**The person affected has the final word on whether their problem was solved**
Status change	FIX_CLAIMED → **RESOLVED** on confirmation, or back to **IN_PROGRESS** on dispute
Exception	No response within the confirmation window → handling is a **policy decision still to be made**; silence is not treated as consent by default

Stage 14 — Resolution or Rework

Confirmed	→ **RESOLVED**, then closed and archived
Not supported by evidence	→ **REWORK / IN_PROGRESS**. The contractor returns, completes the outstanding work, and submits fresh live evidence
Critical rule	**Earlier submissions are never deleted.** The failed submission stays in the record alongside the reason it was not accepted. A record built only from successful submissions is not a record

Stage 15 — Traceable Outcome

The closed case retains, permanently and in a form that cannot be quietly edited:

Evidence

- Original media, unaltered since capture
- Capture hashes, location, timestamps and any `location_unverified / time_unverified` flags
- Work evidence: before, progress and after F

History

- Every status change with actor and time
- Every notification issued
- Dispute and rework history
- Recurrence at this location

Human decisions

- Every acceptance, validation, note, dissent and recusal
- Every institutional decision with its reason F
- The citizen's confirmation or dispute

Integrity

- Append-only, hash-chained audit
- Off-host signed checkpoints
- Discontinuity records instead of chain repair
- No deletion capability anywhere in the system

What the audit layer does and does not prove. It detects change to the recorded history. It does **not** prove that a photograph is truthful or that a decision was correct. Those remain human matters — which is exactly why every consequential step carries a named person and a written reason.

5. UC-01 — Citizen Reporting PHASE 1

Use case ID	UC-01
Actor	Citizen (adult). Students participate only through the supervised route in §12
Objective	Report a real civic problem in a form that another person can later check
Trigger	The citizen observes a problem in public infrastructure
Preconditions	A **verified identity**. **Anonymous reporting is not permitted** — a verified identity is what makes an evidence record credible enough to act on, and what makes the platform resistant to coordinated abuse
Inputs / evidence	Live photographs · GPS location · capture timestamp · a spoken description in **Urdu, English, or a mix of the two**
Primary flow	Capture → confirm location → describe by voice → review the transcript and correct it if needed → submit → receive a case reference → track
System response	Evidence sealed at capture with a content hash; write-once storage; a case reference issued; the report enters the workflow
AI role	Transcription · category suggestion · severity suggestion · possible-duplicate proposal · structured observations. **Advisory, with confidence and model version attached**
Human decision point	A **moderator** decides whether the report enters the workflow. Later, a **representative** decides whether the condition is real
Privacy / safety rule	Identity is verified but held in a **separate encrypted vault** and **never shown** to representatives, officers, contractors, operations staff or the public. Nobody should put themselves at risk to capture evidence — a dangerous problem can be described from a safe distance
Outcome	A traceable, evidence-backed case the citizen can follow, and on which they hold the final confirmation
Alternative flows	**Offline** — capture works fully offline; the bundle is encrypted on the device and uploads on reconnection, with **capture time** governing. **Weak GPS** — manual pin placement, flagged `location_unverified`, still accepted. **Duplicate** — a human decides; if merged, the citizen becomes a supporter and is not accused of anything. **Request for more information** — a specific missing detail is requested, not a vague "more information"
Exception / safety paths	Unsafe location → do not go; describe from a distance. Safeguarding concern in the content → restricted queue, immediate escalation, does not appear on ordinary screens
Audit record	Media captured *(with hash)* · report submitted *(with bundle hash)* · moderation decision *(actor, time, reason)* · every status change · every notification · confirmation or dispute
Phase	PHASE 1

Two rules citizens most often ask about.

Question	Answer
Can I upload a photo I took yesterday?	Not on the live-evidence reporting path. A gallery image could be any image from any place at any time; a live capture, hashed at the moment it is taken with location and time attached, is a claim about a specific place at a specific moment. Only the second kind of image can support an accountability decision
Will anyone know it was me?	Not in normal operation. Identity is removed **in the query that builds every operational view**, so there is no screen on which a reviewer, officer or contractor could see it. Any access to the vault is separately authorised, requires a stated justification, and is logged

6. UC-02 — Street / Community Representative PHASE 1

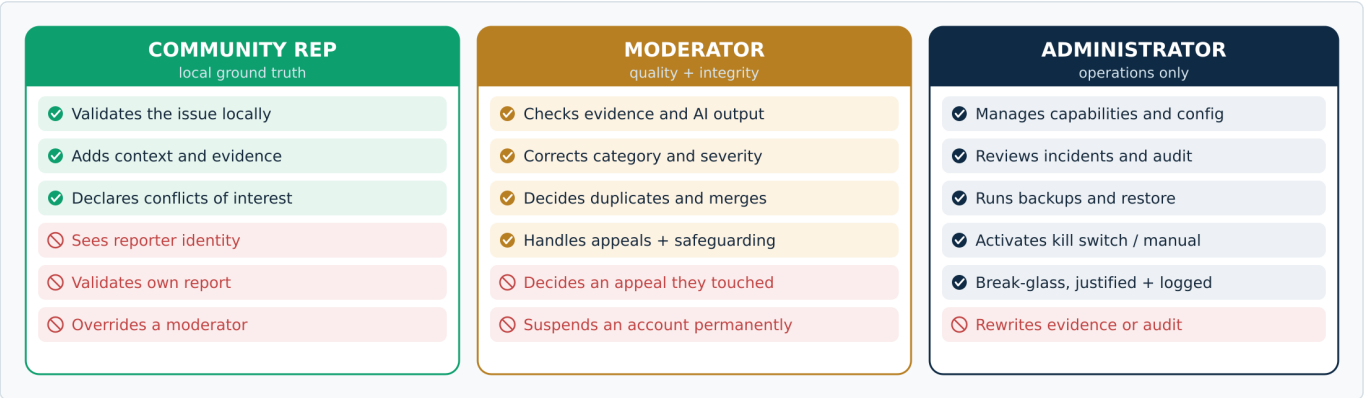


Figure 6.1 — The three Phase-1 human review roles and the boundary of each.

Use case ID	UC-02
Actor	Street / Community Representative — a **consented adult** endorsed by the pilot partner, holding a scoped, time-bounded capability grant. **Students are never representatives**
Objective	Independently verify conditions on the ground, so that a report becomes a verified issue
Trigger	An accepted report is assigned to their geographic scope
Preconditions	Verified adult identity · institutional endorsement · a signed conflict-of-interest declaration, re-attested periodically · training in evidence standards, privacy and safeguarding basics
Inputs / evidence	The evidence packet — original media, transcript, AI annotations with confidence, location, duplicate and related history
Primary flow	Receive a scoped assignment → review the packet → **visit the location where it is safe to do so** → record independent findings with a **required note** → attach supplementary live-captured evidence if it helps → recommend an outcome
System response	Outcome recorded; a **moderator** closes the state; citizen and supporters are notified
AI role	Advisory only, and clearly separated from the original evidence. Where confidence is low the packet says so — a signal to look harder, not a reason to reject
Human decision point	**VALIDATED** or **NOT_VALIDATED**, with a note that says what was actually checked. They may corroborate, challenge, or request one specific missing detail
Privacy / safety rule	**Reporter identity is never visible.** If the representative believes they know who reported something and that belief affects their judgement, they must **recuse**. Nobody should put themselves at risk — an unreachable site is marked `unsafe_to_verify`, which raises severity rather than hiding the issue
Outcome	A recommendation that converts *reported* into *verified*, with a written record of the basis for it

Alternative / exception flows	**Conflict** → recuse; the item returns to the pool and is reassigned; the recusal is logged and counts in the representative's favour. **Outside scope** → reroute suggestion to a moderator. **Already fixed** → record what was found; final handling is a workflow detail still to be confirmed. **Suspected malicious report** → flag to a moderator with what was observed; **no AI model may label a citizen fraudulent**, and a human decides. **Safeguarding concern** → immediate escalation on the restricted route; safeguarding outranks workflow
Audit record	Assignment · every validation, note, supplementary photograph, recusal and dissent, each with actor, time and reason. A sample of validations is independently audited; disagreements trigger review
Phase	PHASE 1

The representative verifies the condition, not the citizen's identity. Two structural consequences follow. First, a representative who validates everything provides no signal at all — challenging a report is as valuable as confirming one. Second, because they recommend rather than decide, a single representative can never close a case alone, and can never make it disappear: **there is no deletion capability in the system**.

Handoff. Once an issue is VERIFIED it becomes actionable. In **Phase 1** the action owner is the pilot-site authority, with updates recorded by proxy by an authorised moderator under audit. In the **future institutional model** the same verified issue reaches a Union Council Officer without the workflow changing — only the capability assignment changes.

7. UC-03 — Union Council Officer / UC Operations

FUTURE INSTITUTIONAL CAPABILITY

Not built. Not deployed. This describes the intended institutional operating model. In Phase 1 there is no formal government authority involved. The statutory mapping between issue categories and the responsible tier or department **must be confirmed with the partner institution and legal review** before any institutional deployment.



Figure 7.1 — The institutional flow, with the contractor-selection and financial boundaries stated separately.

Use case ID	UC-03
Actor	Union Council Officer / UC Operations staff, under a scoped, time-bounded grant with multi-factor authentication
Objective	Convert a verified civic issue into a recorded institutional action
Trigger	An issue reaches **VERIFIED** within the officer's jurisdiction
Preconditions	Authenticated institutional access · jurisdiction scope · **only verified issues reach this queue** — unvalidated reports, rejections and safeguarding-flagged content never appear here
Inputs / evidence	Original evidence · capture metadata and flags · human verification history and the representative's note · verified supporter count · duplicate and recurrence history · full audit trail. **Not reporter identity**

Primary flow	Review the evidence bundle → determine responsibility → accept into the operations workflow **or** reroute with a recorded reason → assign an action owner → record scope, cost, budget reference, timeline and milestones → link the authorised contractor assignment made by the procurement authority
System response	Status moves toward **IN_PROGRESS** on acknowledgement or work start; citizen, supporters and representative are notified
AI role	Anomaly flagging on cost outliers and duplicate work orders. **A flag is a prompt to look, never an accusation and never an automatic rejection**
Human decision point	Every one: responsibility, acceptance, assignment, scope, cost, timeline
Privacy / safety rule	Reporter identity is protected here as everywhere. Field inspection, where an Inspector role is activated, uses the **same tamper-evident capture pipeline citizens use**, and an inspector may not verify works executed by an entity with which they hold a declared relationship
Outcome	A verified issue becomes an actionable, recorded work item with a named owner and a written rationale
Alternative / exception flows	Not a municipal responsibility → reroute with a recorded reason. Wrong operational unit → reassign; **the correction stays visible in the audit trail**. Work stalls → ageing surfaces in dashboards. Dispute after resolution → the issue reopens for rework
Audit record	Officer decision with actor, time and reasoning · assignment and any reassignment · work-order record · every status change
Phase	FUTURE

Parakh workflow is not statutory authority. Parakh records what an authorised official decided and why. It confers no legal power, replaces no procurement process, and does not determine which office is legally responsible for a given category of work — that mapping belongs to the institution and its legal advisers.

8. UC-04 — Contractor / Thekedar FUTURE INSTITUTIONAL CAPABILITY

Not built. Not deployed. Relevant only when Parakh operates alongside a formal procurement and works process. **Parakh does not select contractors, approve contracts, hold money, transfer money, or enforce penalties.**

Use case ID	UC-04
Actor	Contractor entity, acting through **named individual operators** — every action is attributed to a person, not to a faceless company
Objective	Perform authorised work and prove it with evidence comparable to the original
Trigger	An authorised work order is assigned by the procurement authority and recorded or linked in Parakh
Preconditions	Verified organisational identity linked to the entity named in the work order · multi-factor authentication · scope limited to **their own** work orders
Inputs / evidence	Work order: scope, location, start and target completion dates, milestones, approved cost and budget reference, quality commitment, defect liability terms — **and the before evidence**, because completion evidence must be comparable to it
Primary flow	Accept the assignment → record before-work site condition → record progress at each milestone → record any delay **with a written explanation** → raise a variation request where the work as found differs from the work as scoped → submit completion evidence
System response	Automatic checks on live capture, GPS proximity, timestamp window, viewing-angle similarity and integrity hash. A failed check produces a **flag for human review**, not an automatic rejection

AI role	Before/after comparison — describes observable differences and flags apparent mismatches. **Advisory**
Human decision point	A human decides whether the work is acceptable. Variations are decided by the authorising officer. **The contractor can never verify their own work**
Privacy / safety rule	No reporter identity. No visibility of other contractors' work or jurisdiction-wide data
Outcome	**FIX_CLAIMED** — a claim, awaiting independent verification and citizen confirmation
Alternative / exception flows	**Rejected submission** → rework: return to site, complete the outstanding work, submit fresh live evidence. **The earlier submission is never deleted**; it stays with the reason it was not accepted, and the contractor may attach a documented response. **Different defect visible** → recorded as a new observation; a named human decides whether it becomes a new issue or a variation
Audit record	Acceptance · every evidence submission with the named person, time, device attestation and all check results · every delay, explanation and variation request · every rejection and response
Phase	FUTURE

8.1 Defect liability and recurrence

Element	Statement
Defect liability period (DLP)	**Default 12 months**, pending legal review and confirmation in the applicable work order or contract. The contracting authority may specify a different period
Effect	Failures attributable to poor execution during the liability period may require **contractor-funded correction under the applicable contract**
What Parakh does	Detects and links a recurrence at the same location; records the finding and the evidence behind it; makes both available to the authority
What Parakh does not do	**Legally enforce contractual liability.** Enforcement is contractual and belongs to the authority. Parakh supplies the evidence; the authority applies the consequence
Judgement	**A human decides** whether a recurrence is an execution failure, a new and unrelated problem, or ordinary wear. No model, score or automatic rule assigns blame

Performance record. Built from verified events only — work completed and verified, time against commitment, recorded delays and their explanations, evidence rejections and reasons, defect claims and their resolution, recurrence at previously fixed sites. Never subjective opinion, unverified allegation, or any AI-generated character judgement. Any negative finding can be answered with a documented response. *Whether these records are public, authority-only or tiered is a policy decision still to be taken with the partner institution and legal review.*

9. UC-05 — Mayor & Operations Command FUTURE INSTITUTIONAL CAPABILITY

> **Not built as an executive command view.** In Phase 1 a **limited administrative form** exists for the pilot's own operations team. The full jurisdiction-wide, multi-department view is a future capability.

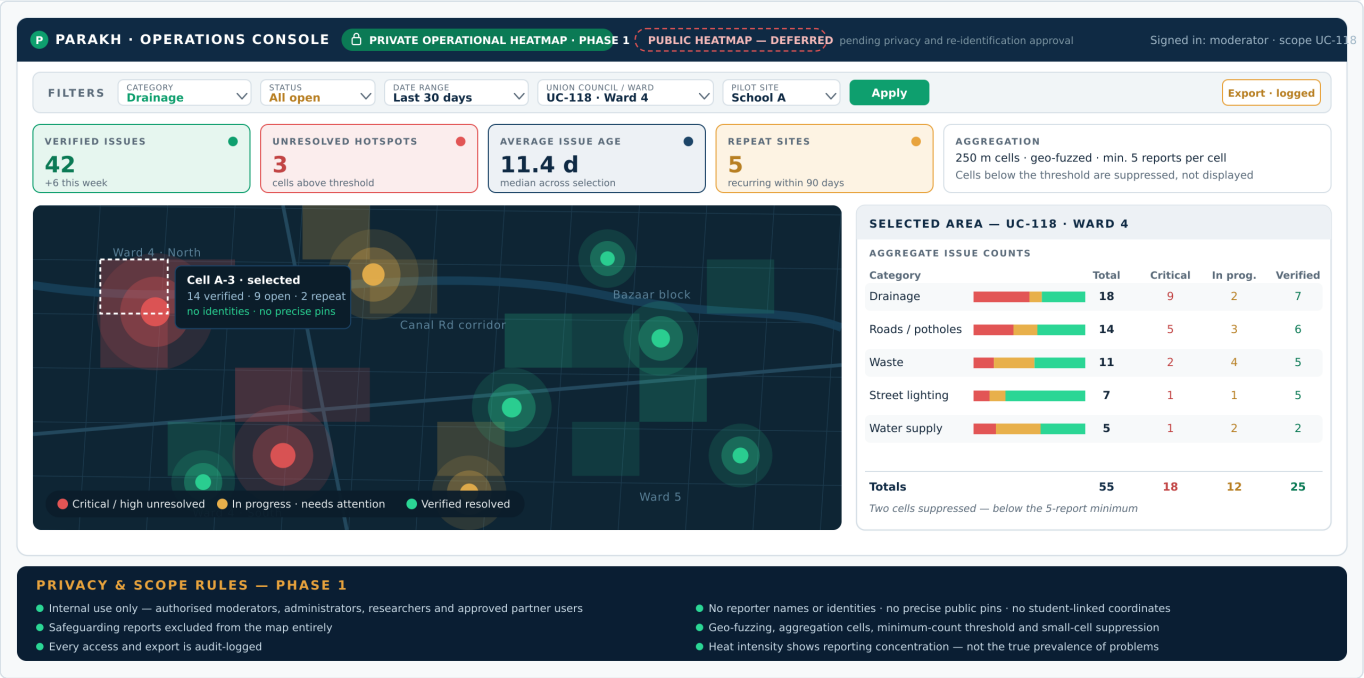


Figure 9.1 — The Phase-1 operational heatmap: private, internal, role-restricted, and deliberately not a public map.

Use case ID	UC-05
Actor	Mayor, senior operations leadership, and authorised operations staff
Objective	**Situational awareness and capacity direction — not individual complaint handling**
Trigger	Routine operational review, or an escalation
Inputs	Aggregated, geo-fuzzed case data · state counts · ageing against benchmarks · category trends · recurrence hotspots · verification capacity and latency · platform-health indicators · work-order visibility where authorised
Primary flow	Read ageing first → read the heatmap for concentration → check recurrence hotspots → check verification capacity → check platform health → direct capacity, escalate, and record the reason
Operational states	**RED** unresolved or critical · **YELLOW** ongoing · **GREEN** verified or resolved
AI role	Anomaly flags and trend surfacing only. **There is no composite AI "health score"** grading a ward, an officer or a contractor — composite scores hide their assumptions, cannot be argued with, and invite decisions nobody can explain afterwards
Human decision point	Where to direct capacity; what to escalate; when a pattern warrants investigation
Privacy / safety rule	Reporter identity is never visible. Every drill-down into case-level data is **logged with actor, time and purpose** — including the Mayor's
Outcome	Operational decisions grounded in inspectable counts rather than a summary number
Phase	FUTURE — limited administrative form in Phase 1

9.1 The heatmap — what it is, and what it is not

The Phase-1 heatmap is private, operational, role-restricted, aggregated and geo-fuzzed, with sensitive and student locations specifically protected. It is not a public surveillance map, and there is no public map in Phase 1.

A public live heatmap is **deferred** behind a formal privacy and re-identification review. Publishing a map of "problem areas" attaches stigma to neighbourhoods, and fine-grained public location data can expose reporters in small communities — which would break the identity protection the whole design rests on.

The heatmap shows reporting patterns, not a perfect map of reality. Three consequences follow, and every user of this portal should hold all three:

- **More reports usually means more trust**, or better coverage — not more decay. A ward whose report count triples after a school partnership has not deteriorated; it has started speaking.
- **A quiet area is a question, not an achievement.** Silence can mean no problems — or no coverage, no connectivity, no trust, or fear.
- **Never make a case count a target.** The moment it becomes one, the incentive shifts from fixing problems to suppressing reports.

10. UC-06 — Human Moderation PHASE 1 — GOVERNANCE LAYER

Use case ID	UC-06
Actor	Moderator *(with administrator for the highest-authority actions)*
Objective	Act as the **human control layer** through which every consequential decision passes
Trigger	A report completes AI processing; a validation outcome is recommended; an appeal, flag or safeguarding concern is raised
Inputs	Original media · transcript and AI annotations, **shown separately and attributed to model and version** · location · duplicate candidates · representative recommendations
Primary flow	Review evidence → accept, reject with a recorded reason, or reroute → decide suspected duplicates → close validation states → handle appeals → apply a temporary restriction where authorised → escalate safeguarding
AI role	Suggestion only. **No AI output may reject, restrict, escalate or label a person**
Human decision point	All of them. Every decision carries the moderator's identity, the time and a written reason
Authority boundary	A moderator may apply a **temporary account restriction only**. **Permanent suspension is an administrator action.** Either way the reason is recorded and the decision is appealable
Privacy / safety rule	Reporter identity is not visible in normal moderation work. Vault access is separately authorised, requires a stated justification, and is logged. Safeguarding concerns go to a restricted queue and are escalated immediately — this **overrides all normal workflow**
Cannot	**Delete a report. There is no such capability**, for moderators, administrators or anyone else. Pressure to make a report disappear cannot be complied with
Outcome	A workflow in which every accepted, rejected, merged, verified or closed case has a named human behind it
Audit record	Every decision, reason, restriction, appeal outcome, escalation and vault access
Phase	PHASE 1

AI assists. Humans decide.

11. Cross-Cutting — Privacy and Reporter Protection

Privacy in Parakh is not a setting. It is a property of how each view is built.

Stage of the case	What is true about identity
Reporting	Identity is **verified**. **Anonymous reporting is not permitted** — verification is what makes an evidence record credible and the platform resistant to coordinated abuse
Storage	Identity sits in a **separate encrypted vault**, isolated from the operational database. Operational records are linked by a pseudonymous participant identifier
AI processing	AI works on evidence and produces annotations stored separately. It never sees, infers or reveals identity
Moderation	The moderator sees the case, not the person
Field verification	The representative **never** sees reporter identity — not in the packet, not in the history, not in an area summary, not in a notification. It is removed **in the query that builds the packet**, not hidden by a screen setting
Institutional action F	The officer sees the evidence bundle without identity
Contractor F	Sees the work order and the before evidence. No identity
Operations F	Aggregated and geo-fuzzed patterns only. No identity, no precise reporter locations
Public	Nothing individual
Any exception	Access to protected identity is **reason-based** — it requires a stated justification recorded at the moment of access — and it is **logged in the append-only audit for every role, including administrators**

11.1 Why this matters operationally

The platform is designed on the assumption that **someone, somewhere, will want a report to disappear or will want to know who made it**. Its structural answers:

- **Identity withheld** from everyone who could act on it.
- **No deletion capability** — not for moderators, not for administrators, not for officers.
- **Append-only, hash-chained audit** with off-host signed checkpoints, so tampering is *detectable* rather than merely prohibited.
- **Every rejection reasoned, named, timed and appealable.**
- **Recusal** as a normal, logged, encouraged act.
- **No public heatmap**, so no neighbourhood is publicly labelled and no reporter is locatable by inference.
- **Sensitive locations** — schools, shelters, health facilities and any site flagged by the pilot partner — receive additional protection.

12. Minor / Student Participation PHASE 1 — SUPERVISED ONLY

Actor	Student under 18, participating **only** through an approved, supervised institutional programme
Objective	Allow young people to contribute civic observations safely, with an adult in the loop
Preconditions	Institutional attestation of the participant · school and guardian consent as required · a named school liaison
Identity	**No CNIC is required from a minor.** Where national identity credentials are not lawfully available to or appropriate for a person under 18, the **school-approved identity and consent route** is used instead: the institution attests the participant, consent is recorded, and only the minimum data needed is stored. A minor is never asked for a credential they cannot lawfully hold
Supervision	**Adult in the loop** throughout. Data minimisation applies more strictly to minors than to anyone else
Hard limits	**Students never become Street Representatives** and never validate anyone else's report. **No minor is ever asked to enter an unsafe location to collect evidence**

Safeguarding	Any welfare concern follows the **restricted safeguarding workflow** — a restricted queue, immediate escalation to an administrator and the named school liaison, and content restriction. **Safeguarding overrides ordinary workflow, always.** Parakh staff do not investigate welfare concerns themselves; they route them to the institution responsible
Withdrawal	Consent can be withdrawn at any time; the account is deactivated, the device queue is rendered unreadable by destroying its key, and the data is excluded from future model training
Phase	PHASE 1, under the approved minors-safe participation model. Full consent, supervision and data-minimisation protocol to be agreed with the pilot partner before launch

13. AI Versus Human Decision



Figure 13.1 — The line between AI assistance and human decision. It does not move.

AI MAY ASSIST WITH

- Transcription of **Urdu and English**, including **Urdu-English code-switching** *(the approved launch scope; Punjabi is a planned extension, not a Phase-1 commitment)*
- Category suggestions
- Severity suggestions
- Possible duplicate detection
- Structured observations from evidence
- Before/After comparison — describing observable differences F
- Recurrence link suggestions
- Cost-outlier and duplicate-work-order flags F
- Confidence values and model/version metadata on everything it produces

AI MUST NOT

- Decide whether a civic complaint is true
- Reject a citizen or a report
- Label a citizen or contractor fraudulent
- Make the final severity or category decision where it is consequential
- Verify that work is complete
- Close a case or change a state
- Approve, hold or release a payment
- Select a contractor
- See, infer or reveal reporter identity
- Modify original evidence
- Rewrite or alter the audit history

13.1 Where the humans decide

Decision point	Who decides	Recorded
Does this report enter the workflow?	Moderator	Actor, time, reason
Is this a duplicate?	Moderator	Merge decision and reason

Decision point	Who decides	Recorded
Does the problem actually exist?	Street Representative *(recommends)* → Moderator *(closes)*	Required note, outcome, time
What is the real severity?	Representative and moderator, informed by local context	Note and any override of the AI suggestion
Is this a municipal responsibility? F	UC Officer	Decision and reasoning
Who is the action owner? F	UC Officer	Assignment and any correction
Which contractor? F	**The procurement authority — outside Parakh**	Authorised assignment recorded or linked
Is the work acceptable?	Independent human reviewer	Outcome, reason, evidence reviewed
Was my problem solved?	**The citizen**	Confirmation or dispute
Is a recurrence a defect? F	Human reviewer, then the authority	Finding and evidence
Any account restriction	Moderator *(temporary)* / Administrator *(permanent)*	Reason; appealable

Why the boundary sits here. Accountability needs a name — a decision affecting someone's street, reputation or livelihood must be attributable to a person who can be asked to explain it, and a model cannot be asked. Models also fail unevenly, and placing consequences behind those failures would place them on the people already least well served. And a platform whose decisions can be dismissed as *"the computer decided"* has nothing to offer a community that already suspects nobody is listening.

14. Rework and Recurrence FUTURE

14.1 Rework — when completion evidence is not accepted

Contractor claims completion → independent verification fails *(wrong location · different angle · outside the work window · repair visibly incomplete · a different defect visible · integrity check failed)* **→ reason recorded, with the reviewer's name and time → the original completion submission remains in the history → contractor performs rework → fresh live evidence submitted → re-verification → outcome recorded**

Rule	Why
The failed submission is **never deleted**	A performance record built only from successful submissions is not a record. The rejections are the signal
The contractor may attach a **documented response**	A finding that cannot be answered is not a fair finding
A failed automated check produces a **flag**, not an automatic rejection	The checks are aids to a human reviewer, not the reviewer

14.2 Recurrence during the defect liability period

Resolved work → the same defect is reported again at the same location → Parakh links the new evidence to the closed case → a human determines: execution failure · new and unrelated problem · ordinary wear → contractual responsibility handled by the authorised institution

Recurrence is not automatically the contractor's fault. A road can fail again for reasons unrelated to the repair — a burst main beneath it, an unusual flood, heavy vehicles it was never specified for. AI may *propose* the link; **a human decides** what it means. And Parakh records the finding — it does not enforce the contract.

15. Appeal and Dispute

Citizen disagrees *(a report was rejected, or a claimed resolution is not a resolution)* → **appeal or dispute recorded, with a reason and optional fresh live-captured evidence → re-reviewed by someone who was not involved in the original decision → reasoned human decision, named and timed → outcome notified → full history preserved**

Situation	Route
Report rejected at moderation	Appeal; re-reviewed by a different person; the original decision and its reason remain visible
Claimed fix disputed by the citizen	Issue returns to **IN_PROGRESS** for rework — this is rework, not argument
Representative disagrees with a moderator	Dissent recorded, visible to administrators, feeding calibration
Contractor disputes a rejection or defect claim F	Documented response attached; escalation through the authority's own appeal route
Account restriction applied	Appealable. Moderator applies temporary restrictions only; permanent suspension is an administrator action

No silent deletion, anywhere. Every decision — including every decision that was later overturned — remains in the append-only history with the name of the person who made it, the time, and the stated reason. The correction is part of the record, not a replacement for it.

16. Important Exception Flows

The exceptions that most often shape whether the model works in practice. *The full exception-training set lives in the User Training & Operations Manual.*

Exception	Correct response	Human responsibility	Audit outcome
No internet at capture	Capture and describe anyway; the bundle is encrypted on the device and queued	None required — the citizen should not wait for signal	Uploads on reconnection; **capture time governs, not upload time**
Weak or unverified GPS	Place the location manually on the map; the report is flagged `location_unverified` and is **still accepted**	Reviewer weighs the flag when judging the case	Location recorded with its accuracy and verification flag
Unsafe verification site	Representative marks `unsafe_to_verify` and does not attend	Never put yourself at risk	The flag **raises** severity for review rather than hiding the issue
Possible duplicate	AI proposes candidates; **a human decides**	Moderator merges or keeps separate	If merged, the second reporter becomes a supporter — not an accused
Suspected malicious report	Flag to a moderator describing what was observed	**A human decides.** No AI model may label a citizen fraudulent	Any restriction is reasoned, recorded and appealable; moderator = temporary only

Exception	Correct response	Human responsibility	Audit outcome
Safeguarding concern	Immediate escalation on the restricted route; content restricted	Overrides all normal workflow. Parakh staff route, they do not investigate	Escalation to administrator and named school liaison, logged
Reviewer conflict of interest	**Recuse.** The item returns to the pool and is reassigned	Recusal is good practice, not failure	Logged, and counts in the reviewer's favour
Contractor work rejected F	Rework: return to site, submit fresh live evidence	Independent human reviewer decides acceptability	The failed submission and its rejection reason remain in history
Recurring defect F	Link to the closed case; raise a defect claim	A human decides execution failure vs new problem vs wear	Finding recorded and made available to the authority
AI unavailable or uncertain	The case proceeds to human review **without annotations**, or with the low confidence shown	Reviewer looks harder; audio remains the evidence	Throughput drops; correctness does not
Wrong operational authority F	Reroute or reassign with a recorded reason	Officer decides	**The original assignment stays visible** — the correction is part of the record

17. The Commit–Verify–Release Boundary FUTURE / PHASE 3

Nothing in this section is built, deployed or active. It describes a future institutional model requiring a formal partnership, a legal framework and a government financial system. It is documented so that everyone understands the boundary before anyone is tempted to cross it.

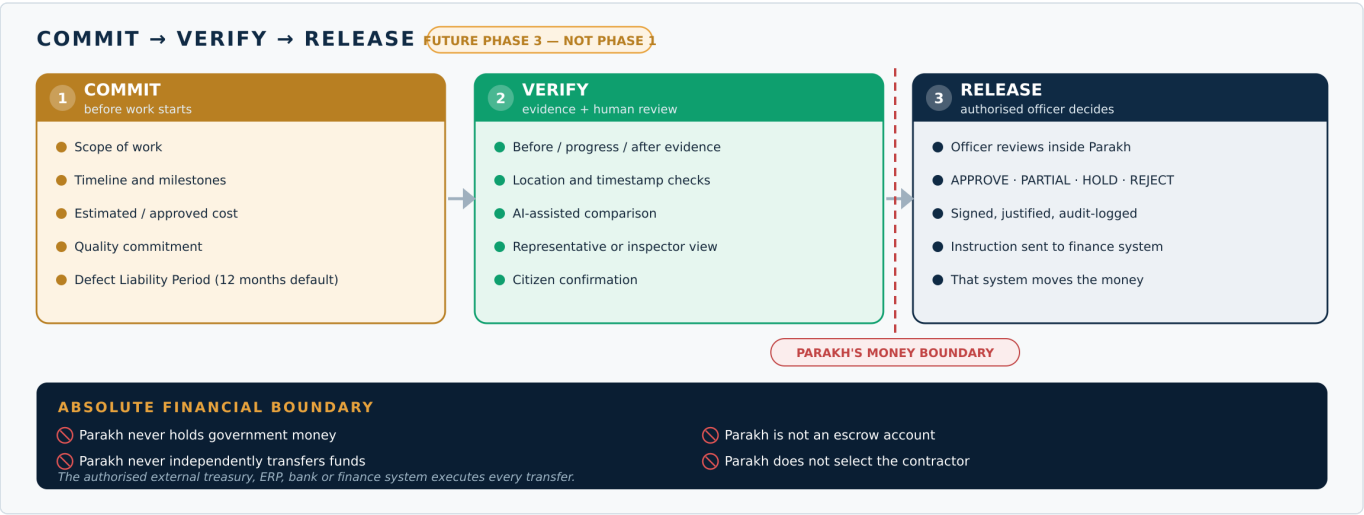


Figure 17.1 — The three stages and the financial boundary.

COMMIT *(authority commits funds against a defined work item, recorded in Parakh)* → **WORK** *(contractor performs the work and evidences it)* → **VERIFY** *(automated checks, then mandatory human review; representative or inspector may corroborate; citizens may confirm)* → **AUTHORISED DECISION** *(an authorised institutional officer records APPROVE · PARTIALLY APPROVE · HOLD · REJECT, with a mandatory reason)* → **EXTERNAL PAYMENT PROCESS** *(the authority's own financial system executes it)*

PARAKH DOES NOT HOLD, POOL, ESCROW, STORE OR INDEPENDENTLY TRANSFER PUBLIC FUNDS.

Parakh is not a bank, an escrow agent, a payment processor, a treasury or a disbursement system. It has no wallet, no balance, no transfer capability and no automated release.

What Parakh does: records a documented officer decision, grounded in verified evidence, with a complete audit trail. **What the authority does:** holds the money and executes the decision, in its own system, under its own controls and its own law.

Four absolute prohibitions: no AI approves a payment · no automatic release by threshold, score, timer or volume of confirmations · no money inside Parakh · **no self-approval** — the person who submits evidence can never be the person who approves against it.

Still to be settled before this could ever be built: legal authority for evidence-linked release · the integration model with the authority's financial system · an appeal process for contractors · liability allocation if verified evidence later proves wrong · independent audit of the decision trail · anti-collusion controls separating officer, inspector and contractor relationships.

18. The End-to-End Accountability Map

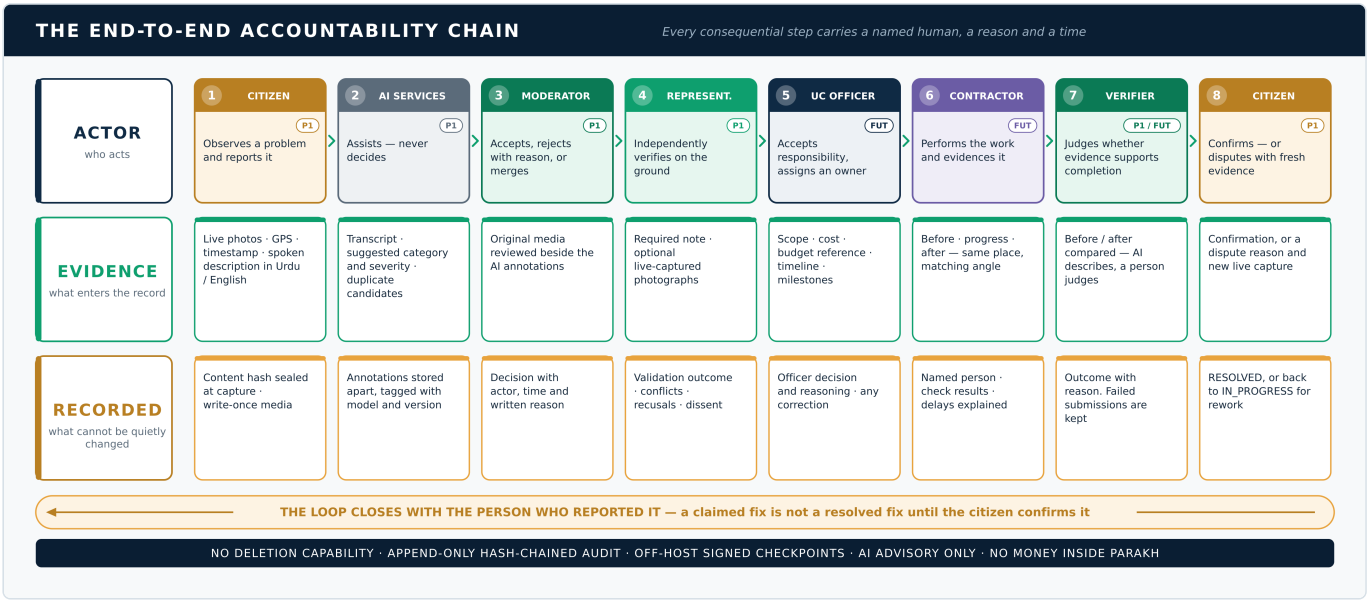


Figure 18.1 — The full chain: who acts, what evidence exists at each point, and what is recorded.

Parakh does not merely record that someone complained,
or that someone marked a task complete.
It creates a traceable chain between observation, evidence, human verification, institutional action, and outcome.

18.1 The chain in one column

Citizen observes → Live evidence captured, sealed at capture → **AI assists**, advisory only → **Human moderates → Representative independently verifies → Institution accepts responsibility** F → **Contractor performs work** F → **Completion evidence submitted** F → **Independent human verifies → Citizen confirms or disputes → Resolved, or rework → Traceable historical record → Aggregated operational insight for authorised leadership** F

18.2 What holds it together

Principle	In practice
Evidence, not assertion	Live capture only on the reporting path; sealed at capture; write-once storage

Principle	In practice
Interpretation kept separate from evidence	AI annotations stored apart from originals, tagged with model and version
A named human at every consequential point	Acceptance, verification, institutional action, work acceptance, closure
The person affected has the final word	A claimed fix is not a resolved fix until the citizen confirms it
Nothing disappears	No deletion capability; append-only, hash-chained audit with off-host signed checkpoints
Clear boundaries	Parakh selects no contractors, holds no money, confers no statutory authority, publishes no map, and lets no model decide

Status, restated. Parakh is currently under design and development. It is **not** an official Government of Punjab application, **no** formal government or NGO partnership has been signed, the complete application is **not yet fully deployed**, and **no social impact has yet been demonstrated**. The Union Council Officer, Contractor, and Mayor & Operations Command portals described here are **future institutional capabilities**.

Where this document and the **Parakh Master Document v1.2.1** disagree, **the Master Document governs**.

Parakh Use Case Document v1.0 · 7 August 2026 · Owner: Hassan — Founder & Project Lead